

A Shared Geometric Object Across the VFD Programme: one object, two roles, an open bridge

(VFD programme — connective note)

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Abstract

This short note records a factual observation and immediately bounds it. The same geometric object—the 600-cell and its icosian ring—appears in two different parts of the VFD programme: *geometrically*, as a substrate in the programme’s cosmological modelling, and *arithmetically*, as the carrier of certain L -functions (companion Paper I). We state plainly that this is a recurrence of one *object* in two *roles*; we make **no claim** that the cosmological model explains, implies, or is explained by the Riemann zeta function. The bridge between the geometric and arithmetic roles is *open*, and for the specific case of the Riemann ζ it has been tested and found *negative* (the (O2) result, companion Paper II).

1 The observation

The 600-cell / icosian ring is a single, well-defined mathematical object. In the programme it is used in two distinct ways:

- **geometric role:** as a 4-dimensional substrate / vertex configuration in cosmological modelling;
- **arithmetic role:** as the order whose theta series realizes $\zeta_K(s)\zeta_K(s-1)$, with sub-objects realizing $\zeta(s)\zeta(s-1)$ and $L(\chi_5)$ (Paper I).

That one object recurs in both is true and worth recording. It is the natural reason the same vocabulary (spheres, polytopes, icosahedral symmetry) appears in both settings.

2 What is *not* claimed

We are explicit, because the tempting overstatement is exactly the one to avoid.

1. We do **not** claim that the cosmological model is, encodes, or explains the Riemann ζ or its zeros.
2. We do **not** claim a proven bridge between the geometric and arithmetic roles of the object. That bridge is *open*.
3. For the specific, strongest form of such a bridge—“the substrate realizes the Riemann ζ ”—the engine of Paper II returns a *certified negative* on five independent grounds: the substrate realizes $\zeta_K = \zeta \cdot L(\chi_5)$, which contains ζ only as a factor.

3 Two cautionary anchors

Two elementary facts guard against reading more into the recurrence than is there.

- **Shared silhouette is not shared content.** Diagrams in cosmology and in analytic number theory can share an outline (e.g. a flaring/funnel shape) while denoting unrelated objects; the visual rhyme carries no implication.
- **Universality is generic.** The Riemann zeros share their fine statistics (GUE) with a vast range of unrelated systems; sharing a universality class is therefore *weak* evidence of any specific identity.

4 Statement of record

One object recurs in two roles; the cross-role bridge is open; the specific ζ -bridge is certified negative; no claim links the cosmological model to the Riemann Hypothesis. This note exists so that the recurrence can be cited accurately without being inflated.